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N-body Simulations of Spiral Bending Waves in Planetary Rings

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ABSTRACT

Draft. This mock paper currently holds the derivation of the vertical epicyclic motion used by the epi_int_3d drift step (Section 2), the vertical gravity that one streamline exerts on another (Section 3), and a study of the streamline method's resolution limit (Appendix B).

1. INTRODUCTION

The epi_int and epi_int_lite N-body integrators (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025) model a planetary ring as a set of gravitating streamlines whose particles are confined to the ring plane. Spiral bending waves, which are launched at a satellite's vertical resonances (F. H. Shu et al. 1983; F. H. Shu 1984), require the particles' vertical motions too. This paper describes epi_int_3d, a modest extension of epi_int_lite into the vertical (out-of-plane) direction, and uses it to simulate spiral bending waves.

2. VERTICAL EPICYCLIC MOTION

The epi_int_lite drift step advances each particle's unperturbed orbit about an oblate planet using first-order epicyclic equations that are exactly reversible: converting spatial coordinates into epicyclic orbit elements and back introduces no error, so the drift step is not a source of numerical drift (J. M. Hahn et al. 2025). Here those equations are extended to particles having vertical displacement z and vertical velocity v_z, with the same first-order accuracy and the same exact reversibility.

2.1. *Equations of motion*

A particle orbiting an oblate planet obeys Newton's second law,

$$\ddot{\mathbf{r}} = -\nabla\Phi, \tag{1}$$

where Φ is the planet's gravitational potential. That potential is a series in the planet's zonal harmonics J_{2k} (Eqn. 5 of N. Borderies-Rappaport & P.-Y. Longaretti 1994), and as in J. M. Hahn & J. N. Spitale (2013) and J. M. Hahn et al. (2025) only the J_2 term is kept, so

$$\Phi(r,z) = -\frac{GM_p}{\rho}\left[1 - J_2\left(\frac{R_p}{\rho}\right)^2 P_2\left(\frac{z}{\rho}\right)\right]. \tag{2}$$

Here M_p is the planet's mass, R_p its radius, r the particle's cylindrical radius, $\rho^2 = r^2 + z^2$, and $P_2(\mu) = (3\mu^2-1)/2$. In cylindrical coordinates (r,θ,z), Eqn. (1) is

$$\begin{aligned} \ddot{r} - r\dot{\theta}^2 &= -\frac{\partial\Phi}{\partial r} \\ \frac{d}{dt}\left(r^2\dot{\theta}\right) &= -\frac{\partial\Phi}{\partial\theta} = 0 \\ \ddot{z} &= -\frac{\partial\Phi}{\partial z}. \end{aligned} \tag{3}$$

The potential is independent of θ, so the z-component of the specific angular momentum, $h = r^2\dot{\theta} = rv_\theta$, is conserved exactly, even when $z \neq 0$. The potential is also even in z, so near the ring plane

$$\Phi(r,z) = \Phi(r,0) + \frac{1}{2}\nu^2(r)z^2 + \mathcal{O}(z^4), \tag{4}$$

where $\Phi(r,0) = -GM_p/r - J_2GM_pR_p^2/2r^3$ is the midplane potential and

$$\nu^2(r) = \left.\frac{\partial^2\Phi}{\partial z^2}\right|_{z=0} = \frac{GM_p}{r^3}\left[1 + \frac{9}{2}J_2\left(\frac{R_p}{r}\right)^2\right]. \tag{5}$$

Eqns. (3) then become

$$\begin{aligned} \ddot{r} &= \frac{h^2}{r^3} - \frac{\partial\Phi(r,0)}{\partial r} + \mathcal{O}(z^2) \\ \ddot{z} &= -\nu^2(r)z + \mathcal{O}(z^3). \end{aligned} \tag{6}$$

To first order the radial equation does not involve z; it is the planar equation of motion. J. M. Hahn et al. (2025) solved the radial part of Eqns. (6) to first order

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in small displacements from circular motion, in a form that is exactly reversible: the conversion from the coordinates r, θ, v_r, v_θ to the epicyclic elements $a, e, \tilde{\omega}, M$, and back, is exact. This is required if the solution is to be used in the drift step of a drift-kick N-body integrator. That solution is summarized in Appendix A. In it the particle's radius is $r = a[1 + \mathcal{O}(e)]$, its mean longitude $\lambda = \tilde{\omega} + (\Omega_a/\kappa_a)M$ advances at the mean angular velocity Ω_a, and its mean anomaly M advances at the epicyclic frequency κ_a. Note that Ω denotes the longitude of ascending node, while Ω_a is the mean angular velocity.

So the only additional physics that epi_int_lite's drift step needs in order to follow out-of-plane motion is the vertical equation of motion, the second of Eqns. (6); the vertical forces that the ring and satellites exert enter the kick step and are treated separately. Evaluating $\nu^2(r)$ at $r = a$ reduces the vertical equation to the harmonic oscillator

$$\ddot{z} = -\nu_a^2 z, \tag{7}$$

with vertical frequency (Eqn. 17 of N. Borderies-Rappaport & P.-Y. Longaretti 1994)

$$\nu_a \equiv \nu(a) = \sqrt{\frac{GM_p}{a^3}}\left[1 + \frac{9}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}, \tag{8}$$

where the neglected terms are of order $\mathcal{O}(ei)$, due to the particle's radial excursion $r - a = \mathcal{O}(ea)$, and $\mathcal{O}(i^3)$. Outside the planet the potential obeys Laplace's equation, which for an axisymmetric potential is

$$\nabla^2\Phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\Phi}{\partial r}\right) + \frac{\partial^2\Phi}{\partial z^2} = 0. \tag{9}$$

At $r = a$ and $z = 0$, where $\Omega_a^2 = r^{-1}\partial\Phi/\partial r$, $\kappa_a^2 = 3\Omega_a^2 + \partial^2\Phi/\partial r^2$ and $\nu_a^2 = \partial^2\Phi/\partial z^2$, Eqn. (9) relates the three frequencies via $\kappa_a^2 + \nu_a^2 = 2\Omega_a^2$.

2.2. *Vertical solution*

The solution of Eqn. (7) is a sinusoid of frequency ν_a whose amplitude and phase define two new orbit elements, the inclination i and the longitude of ascending node Ω:

$$\begin{aligned} z &= a\, i \sin(\lambda - \Omega) \\ v_z &= a\nu_a i \cos(\lambda - \Omega), \end{aligned} \tag{10}$$

where the node regresses at the rate

$$\dot{\Omega} = \Omega_a - \nu_a \simeq -\frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\Omega_a. \tag{11}$$

The vertical phase $\lambda - \Omega$ then advances at exactly ν_a, so Eqns. (10) solve Eqn. (7) exactly, with $v_z = \dot{z}$, and the particle crosses the ring plane upward at longitude $\lambda = \Omega$. Measuring the phase from λ rather than θ is what keeps the vertical motion a pure harmonic oscillator; $\theta - \Omega$ would add an $\mathcal{O}(ei)$ error to the phase.

2.3. *Coordinates to elements*

Eqns. (10) are inverted by a fifth step that follows steps 1–4 of Appendix A. Step 1 remains exact in three dimensions because h is conserved and r is the cylindrical radius, so steps 1–4 provide a, Ω_a, κ_a, e, M, $\tilde{\omega}$, and so λ. Step 5 computes ν_a from Eqn. (8) and then

$$\begin{aligned} i &= \sqrt{\left(\frac{z}{a}\right)^2 + \left(\frac{v_z}{a\nu_a}\right)^2} \\ \Omega &= \lambda - \mathrm{atan2}(z, v_z/\nu_a). \end{aligned} \tag{12}$$

Once a is known, the mapping $(i, \lambda - \Omega) \leftrightarrow (z, v_z/\nu_a)$ is a change between polar and Cartesian coordinates, so Eqns. (10) and (12) are exact inverses. The drift step from t to $t + \Delta t$ converts coordinates to elements, advances $M \to M + \kappa_a\Delta t$ and $\Omega \to \Omega + (\Omega_a - \nu_a)\Delta t$, and converts back. Its vertical part is simply a rotation of $(z, v_z/\nu_a)$ through the angle $\nu_a\Delta t$.

3. THE VERTICAL GRAVITY OF A STREAMLINE

In epi_int and epi_int_lite (Section 2.3.2 of J. M. Hahn & J. N. Spitale 2013; Eqn. 1 of J. M. Hahn et al. 2025), the gravity that a perturbing streamline a' exerts on a particle in another streamline is that of a straight and infinitely long wire of linear density λ, which pulls the particle toward it with

$$A_g = \frac{2G\lambda}{\Delta}, \tag{13}$$

where Δ is the particle's perpendicular distance from the wire (Eqn. 19 of J. M. Hahn & J. N. Spitale 2013). Figure 1a redraws Figure 3 of J. M. Hahn & J. N. Spitale (2013). The wire passes through the point where streamline a' crosses the particle's longitude, a distance $\Delta_r = r' - r$ outward, which is found by fitting a Lagrange polynomial to the three particles of a' nearest in longitude. The wire runs along the streamline's velocity $\mathbf{v}'$ there, also interpolated, at the angle φ to $\hat{\theta}$, where $\sin\varphi = v_r'/v'$ and $\cos\varphi = v_\theta'/v'$. The perpendicular from the particle to the wire then makes the same angle φ with $\hat{r}$, so A_g has the components

$$\begin{aligned} A_{g,r} &= A_g\cos\varphi \\ A_{g,\theta} &= -A_g\sin\varphi, \end{aligned} \tag{14}$$

which J. M. Hahn & J. N. Spitale (2013) approximate as $A_{g,r} \simeq A_g$ and $A_{g,\theta} \simeq -A_g v_r'/v_\theta'$ to lowest order in the perturbing streamline's eccentricity e' (their Eqns. 20).

3.1. *An inclined wire*

In three dimensions the perturbing streamline also crosses the particle's longitude a height $\Delta_z = z' - z$ above the particle, with z' interpolated in the same way as r', and its velocity there has a vertical component v'_z. Eqn. (13) still holds, with the pull directed along the perpendicular $\mathbf{\Delta} = (\Delta_r, \Delta_\theta, \Delta_z)$ from the particle to the wire, so that

$$\begin{aligned} A_{g,r} &= A_g \Delta_r / \Delta \\ A_{g,\theta} &= A_g \Delta_\theta / \Delta \\ A_{g,z} &= A_g \Delta_z / \Delta. \end{aligned} \tag{15}$$

The Lagrange polynomial provides the crossing point rather than the nearest point of the wire, so the perpendicular follows from the separation $\mathbf{d} = (r' - r, 0, z' - z)$ from the particle to the crossing point and the wire's direction $\hat{\mathbf{v}}' = \mathbf{v}'/v'$. Points on the wire lie at $\mathbf{d} + s\hat{\mathbf{v}}'$ from the particle, where s is the distance along the wire from the crossing point, and the nearest of them is where that separation is normal to the wire, $(\mathbf{d} + s\hat{\mathbf{v}}') \cdot \hat{\mathbf{v}}' = 0$, i.e. at $s = -\mathbf{d} \cdot \hat{\mathbf{v}}'$. So

$$\mathbf{\Delta} = \mathbf{d} - (\mathbf{d} \cdot \hat{\mathbf{v}}')\hat{\mathbf{v}}' = \hat{\mathbf{v}}' \times (\mathbf{d} \times \hat{\mathbf{v}}'), \tag{16}$$

with all vectors expressed in the local frame $(\hat{r}, \hat{\theta}, \hat{z})$, which is the same at the particle and at the crossing point since they share a longitude. Eqns. (13), (15) and (16), i.e. $\mathbf{A}_g = 2G\lambda\mathbf{\Delta}/\Delta^2$, are exact for a straight wire. The component of $\mathbf{d}$ along the wire, $\mathbf{d} \cdot \hat{\mathbf{v}}' = [(r' - r)v'_r + (z' - z)v'_z]/v'$, is first order in e' and the inclination i', so to first order

$$\begin{aligned} \Delta_r &= r' - r, \qquad \Delta_z = z' - z, \\ \Delta_\theta &= -\Delta_r \sin\varphi - \Delta_z \sin\chi, \\ \Delta &= (\Delta_r^2 + \Delta_z^2)^{1/2}, \end{aligned} \tag{17}$$

where $\sin\chi = v'_z/v'$. Eqns. (13), (15) and (17) are thus accurate to first order in the perturbing streamline's eccentricity and inclination, with errors of order e'^2, i'^2 and $e'i'$.

The angle χ is the vertical counterpart of φ. Figure 1b is Figure 1a with z in place of r: viewed from outside the ring, the wire is tilted by χ from $\hat{\theta}$, along $\mathbf{v}'$, so the perpendicular leans toward $-\hat{\theta}$ by χ. Figure 1c shows the r–z plane at the particle's longitude (the particle's meridional plane), where the wire crosses at (r', z'), nearly at right angles. When $\Delta_z = 0$, $\Delta_r/\Delta = \cos\varphi$ and $\Delta_\theta/\Delta = -\sin\varphi$, which is Eqn. (14). In that case the planar $\Delta = \Delta_r \cos\varphi$ exactly, which Eqn. (17) approximates by Δ_r.

The vertical acceleration does not require $|\Delta_z| \ll |\Delta_r|$, so it remains valid when neighbouring streamlines are displaced vertically by as much as their spacing. That happens when a bending wave's amplitude approaches its wavelength, and it is where the in-plane and vertical motions couple: there the radial pull $A_{g,r} = 2G\lambda\Delta_r/\Delta^2$ is weaker than the planar $2G\lambda/\Delta_r$ by the factor Δ_r^2/Δ^2.

3.2. *The restoring force of a bending wave*

Eqns. (15) recover the self-gravity of a bending wave. Consider a sheet of surface density σ made of streamlines with $\lambda = \sigma\, dr'$, displaced vertically to $Z(r) = Z_0 \cos kr$. When $|\Delta_z| \ll |\Delta_r|$, the sum of A_z over the sheet is

$$A_z = 2G\sigma Z_0 \int_{-\infty}^{\infty} \frac{\cos kx - 1}{x^2}\, dx = -2\pi G\sigma |k| Z, \tag{18}$$

which is the self-gravity term of the bending-wave dispersion relation, $(\omega - m\Omega_a)^2 = \nu_a^2 + 2\pi G\sigma|k|$ (F. H. Shu et al. 1983; F. H. Shu 1984). So summing the wires' pulls over the streamlines gives the vertical restoring force of a self-gravitating sheet, just as the same sum gives the $-2\pi G\sigma|k|$ term of a density wave in the plane. For bending waves, though, self-gravity is restoring, and adds to ν_a^2.

On the code's lattice of streamlines spaced Δ apart, with $Z_j = Z_0 \cos(j\phi)$ and $\phi = k\Delta$ as in Appendix B,

$$A_z = \frac{4G\sigma Z_0}{\Delta} \sum_{n=1}^{\infty} \frac{\cos n\phi - 1}{n^2} = -X(\phi) Z_0, \tag{19}$$

where $X(\phi)$ is the lattice self-gravity of Eqn. (B3). The vertical lattice modes therefore oscillate at $\omega^2 = \nu_a^2 + X(\phi)$, which tends to the continuum value as $\phi \to 0$. At the zigzag $\phi = \pi$, the extra restoring force is half the continuum value, as it is in the plane. Vertically, however, the lattice's self-gravity is stabilizing, so the resolution limit of Eqn. (B6), which concerns in-plane modes, has no vertical counterpart.

3.3. *The wire's direction*

J. M. Hahn & J. N. Spitale (2013) take the wire's direction from the perturbing streamline's velocity, whereas its gravity comes from the streamline's instantaneous shape. If that shape, $r'(\theta - \Omega_p t)$ and $z'(\theta - \Omega_p t)$, rotates at the pattern speed Ω_p, then

$$\frac{v'_r}{v'_\theta} = \left(1 - \frac{\Omega_p}{\Omega_a}\right) \frac{1}{r} \frac{\partial r'}{\partial \theta}, \tag{20}$$

and likewise for v'_z/v'_θ. For the $m = 1$ eccentric ringlets of J. M. Hahn et al. (2025), Ω_p is the slow precession rate, so the velocity and the shape agree to $\mathcal{O}(J_2)$. For an m-armed pattern near its resonance, the velocity's slope is about $1/m$ times the shape's. epi_int_3d keeps

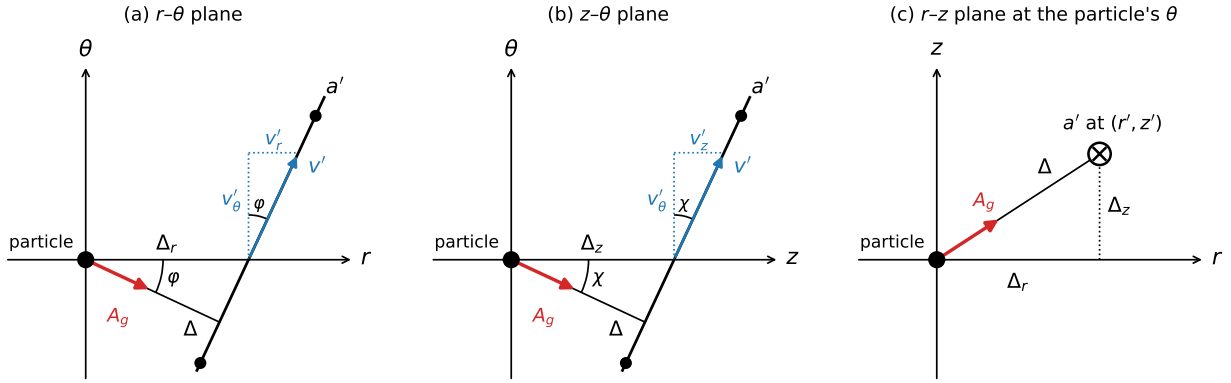


Figure 1. The gravity of a perturbing streamline a', treated as a straight wire, on a particle. (a) The r–θ plane, as in Figure 3 of J. M. Hahn & J. N. Spitale (2013): the wire crosses the particle's longitude a distance Δ_r away and runs along the streamline's velocity $\mathbf{v}'$, tilted by φ from $\hat{\theta}$, so the perpendicular Δ from the particle to the wire, along which it pulls with $A_g = 2G\lambda/\Delta$, makes the angle φ with $\hat{r}$. (b) The same in the z–θ plane, seen from outside the ring: the wire crosses a height Δ_z above the particle, tilted by χ from $\hat{\theta}$ by v'_z. (c) The r–z plane at the particle's longitude, which the wire crosses at (r', z'), running into the page: to first order $\Delta = (\Delta_r^2 + \Delta_z^2)^{1/2}$ (Eqns. 17).

the velocity tangent, which is part of the wire model of J. M. Hahn & J. N. Spitale (2013) and J. M. Hahn et al. (2025). In any case χ enters only A_θ, so the vertical accelerations do not depend on this choice.

3.4. *Implementation*

epi_int_3d is to compute Eqns. (15) in a form that reduces exactly to epi_int_lite's arithmetic when all streamlines lie in the plane. epi_int_lite computes Eqns. (14) with $A_g = 2G\lambda/\Delta_r$, i.e. with Δ_r in place of the planar $\Delta = \Delta_r \cos\varphi$, and with $v' = (v_r'^2 + v_\theta'^2)^{1/2}$; both differ from the exact wire at $\mathcal{O}(e'^2)$. Extended in the same way, with $q = \Delta_z/\Delta_r$,

$$\begin{aligned} A &= \frac{2G\lambda}{\Delta_r(1+q^2)} = \frac{2G\lambda\Delta_r}{\Delta^2} \\ A_{g,r} &= A\cos\varphi \\ A_{g,\theta} &= -A(\sin\varphi + q\sin\chi) \\ A_{g,z} &= Aq, \end{aligned} \tag{21}$$

which is Eqns. (15) and (17) to first order, and where z' and v'_z are interpolated to the particle's longitude in the same way as r' and v'_r. Δ_r never vanishes, since streamlines may not cross. As in epi_int_lite, the particle's own streamline exerts no force on it. In three dimensions that streamline's pull would have no vertical part to lowest order, because the particle lies on it, so $\Delta_z = 0$. Its out-of-plane curvature would add a vertical term of relative size about $m^2\Delta_r/kr$, times a logarithm, which is small when $\Delta_r \ll 2\pi/k \ll r$. epi_int_lite evaluates λ at the perturbed particle rather than on the perturbing streamline. epi_int_3d can instead interpolate the perturbing streamline's own λ' to the particle's longitude, as it does r'. In the eccentric ringlets of J. M. Hahn et al. (2025) this changes the precession rate by about 10^{-7} of itself. The wire approximation requires that Δ be much smaller than the length r/m over which the streamline curves in longitude, as in two dimensions.

APPENDIX

A. THE IN-PLANE SOLUTION OF HAHN ET AL. (2025)

This appendix summarizes the first-order, exactly reversible solution of the planar equation of motion from Appendix A of J. M. Hahn et al. (2025), in the notation used here. The mean angular velocity and epicyclic frequency at semimajor axis a are

$$\Omega_a = \sqrt{\frac{GM_p}{a^3}}\left[1+\frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}$$
$$\kappa_a = \sqrt{\frac{GM_p}{a^3}}\left[1-\frac{3}{2}J_2\left(\frac{R_p}{a}\right)^2\right]^{1/2}. \tag{A1}$$

In terms of the epicyclic elements a, e, $\tilde{\omega}$ and M, the particle's coordinates are

$$r = \frac{a}{\sqrt{1+2e\cos M}}$$
$$\theta = \frac{\Omega_a}{\kappa_a}(M+2e\sin M)+\tilde{\omega}$$
$$v_r = ea\kappa_a \sin M\left(\frac{r}{a}\right)^3$$
$$v_\theta = \frac{a^2\Omega_a}{r}, \tag{A2}$$

which conserve $h = rv_\theta = a^2\Omega_a$ exactly and solve the planar equation of motion with errors of order $\mathcal{O}(e^2)$. The elements advance as $M \to M+\kappa_a\Delta t$, with a, e and $\tilde{\omega}$ constant. Eqns. (A2) are inverted exactly by:

1. Compute $h = rv_\theta$, then solve $h^2 = a^4\Omega_a^2$ for the semimajor axis, $a = R_p\left(C+\sqrt{C^2-3J_2/2}\right)$ where $C = h^2/2GM_pR_p$.
2. Compute Ω_a and κ_a from Eqns. (A1).
3. Compute $e\sin M = (v_r/a\kappa_a)(a/r)^3$ and $e\cos M = \frac{1}{2}[(a/r)^2-1]$, which give e and M.
4. Compute $\tilde{\omega} = \theta - (\Omega_a/\kappa_a)(M+2e\sin M)$.

B. RESOLUTION LIMIT OF THE STREAMLINE METHOD

A streamline model represents a continuous ring by N_r streamlines spaced Δ apart, and that discreteness sets which wavelengths of the ring's response it can follow. This appendix derives the axisymmetric stability of a uniform lattice of circular streamlines under the self-gravity and pressure accelerations of epi_int_lite (J. M. Hahn & J. N. Spitale 2013; J. M. Hahn et al. 2025), and compares it with the continuum Toomre criterion.

Each streamline of linear density λ exerts the acceleration $2G\lambda/d$ of a straight wire on a particle a distance d away (J. M. Hahn & J. N. Spitale 2013). The pressure $p = c^2\sigma$ acts through the surface density $\sigma_j = 2\lambda/(r_{j+1}-r_{j-1})$ of streamline j and the central difference $A_j = -(p_{j+1}-p_{j-1})/2\lambda$, which tends to $-\sigma^{-1}\partial p/\partial r$ as $\Delta \to 0$. Displace the streamlines radially by $\xi_j = \xi\cos(j\phi)$, where $\phi = k\Delta$ and k is the radial wavenumber, so that $0 < \phi \le \pi$; $\phi = \pi$ is the shortest wave the lattice can carry, the zigzag $\xi_j = (-1)^j\xi$ of Figure 2a. Linearizing in ξ, the streamlines at $j = \pm n$ add $4G\lambda(1-\cos n\phi)\xi/(n\Delta)^2$ to the acceleration of streamline 0, and summing over n with $\sigma = \lambda/\Delta$ gives

$$X(\phi) = \frac{4G\sigma}{\Delta}\sum_{n=1}^{\infty}\frac{1-\cos n\phi}{n^2} = \frac{4G\sigma}{\Delta}\left(\frac{\pi\phi}{2}-\frac{\phi^2}{4}\right), \tag{B3}$$

while pressure contributes

$$\omega_p^2(\phi) = \left(\frac{c}{\Delta}\right)^2\sin^2\phi, \tag{B4}$$

so that each mode oscillates at the frequency

$$\omega^2 = \kappa^2 - X(\phi) + \omega_p^2(\phi). \tag{B5}$$

For $\phi \ll 1$ these recover the continuum terms $X \to 2\pi G\sigma k$ and $\omega_p^2 \to c^2k^2$, and Eqn. (B5) becomes the dispersion relation of a pressure-supported self-gravitating sheet, which is stable at all k when $Q = c\kappa/\pi G\sigma \geq 1$ (A. Toomre 1964; J. Binney & S. Tremaine 1987). At the grid scale the lattice departs from the continuum in two ways (Figure 2b). Self-gravity peaks at $X(\pi) = \pi^2 G\sigma/\Delta$, half the continuum value $2\pi G\sigma k$ at $k = \pi/\Delta$. And pressure vanishes there: in the zigzag, streamlines $j \pm 1$ move together, so $r_{j+1} - r_{j-1}$ and hence σ_j and p_j do not change. The viscous fluxes of J. M. Hahn & J. N. Spitale (2013), which are differenced in the same way, likewise exert no force on the zigzag.

A cold continuum disk ($c = 0$) is unstable to all wavelengths shorter than $\lambda_{\rm crit} = 4\pi^2 G\sigma/\kappa^2$, and the lattice hides these: by Eqn. (B3) a cold lattice is axisymmetrically stable as long as

$$\mathcal{R} \equiv \frac{\lambda_{\rm crit}}{2\Delta} = \frac{2\pi^2 G\sigma}{\kappa^2 \Delta} \simeq \frac{\pi G m_r N_r}{\kappa^2 a \Delta a^2} < 2, \tag{B6}$$

where 2Δ is the shortest resolved wavelength, and the last form is for a ringlet of mass m_r, semimajor axis a and width Δa, with $\sigma \simeq m_r/2\pi a\Delta a$ and $\Delta \simeq \Delta a/N_r$. Adding streamlines to a given ringlet raises $\mathcal{R}$ in proportion to N_r.

A viscous ringlet must also avoid the viscous overstability, the growth of axisymmetric epicyclic oscillations that viscous stresses drive when the shear stress rises steeply enough with surface density (U. Schmit & W. M. Tscharnuter 1995; P.-Y. Longaretti & N. Rappaport 1995; H. Salo et al. 2001; J. Schmidt et al. 2001; H. N. Latter & G. I. Ogilvie 2006, 2008). For a cold ringlet with constant kinematic shear and bulk viscosities ν_s and ν_b, as in epi_int_lite, and Keplerian $\kappa = \Omega$, a mode with oscillation frequency $\omega \simeq \kappa(1-x)^{1/2}$ grows to first order in the viscosities at the rate

$$\gamma = \frac{K}{6(1-x)} \left[\nu_s(4x+2) - 3\nu_b(1-x)\right], \tag{B7}$$

where $K = k^2$ and $x = 2\pi G\sigma k/\kappa^2$ in the continuum. Without self-gravity ($x = 0$) the ringlet is stable when $\nu_b > 2\nu_s/3$, the onset found by U. Schmit & W. M. Tscharnuter (1995) and M. Lehmann et al. (2017) for a viscosity independent of σ, while self-gravity is destabilizing (M. Lehmann et al. 2017): a mode is stable only for $x < (3\nu_b - 2\nu_s)/(4\nu_s + 3\nu_b)$, which is 1/7 for $\nu_b = \nu_s$ and 0.4 for $\nu_b = 2\nu_s$. The streamline lattice follows Eqn. (B7) with the grid's versions of both terms: $K = \sin^2\phi/\Delta^2$, from the code's central differences of the viscous stresses, and $x = X(\phi)/\kappa^2$. Both stresses vanish at the zigzag, $\phi = \pi$, where self-gravity is strongest, $X(\pi)/\kappa^2 = \mathcal{R}/2$; so a viscous lattice is stable at every wavelength when $\mathcal{R} < 2/7$ for $\nu_b = \nu_s$, or $\mathcal{R} < 4/5$ for $\nu_b = 2\nu_s$, more stringent than Eqn. (B6). Since $K\nu \sim \nu/\Delta^2$, an overstable lattice's modes grow faster the more streamlines it has.

That limit matters because a ringlet's self-gravity must be strong enough to resist the differential precession due to J_2. Across a ringlet of width Δa, J_2 drives a differential precession $\propto J_2(R_p/a)^2\Omega_a\Delta a/a$, while for a small nonlinearity parameter the ringlet's self-gravity drives one $\propto (m_r/M_p)\Omega_a\Delta\tilde{\omega}(a/\Delta a)^2$, where $\Delta\tilde{\omega}$ is the periapse twist across it (cf. Eqn. C1 of J. M. Hahn et al. 2025). So keeping J_2 from winding up the ringlet requires $m_r \propto J_2\Delta a^3/\Delta\tilde{\omega}$, and then $\mathcal{R} \propto J_2\Delta a N_r/\Delta\tilde{\omega}$: a wider ringlet does not help, and for a given planet more streamlines can be afforded only by a narrower ringlet. We tested this with a ringlet of width $\Delta a = 0.0125a$ whose edges are tapered, mass $m_r = 3\times10^{-8}M_p$, eccentricity $e = 10^{-4}$, periapses twisted by π from its inner to outer edge, and kinematic viscosity $\nu = 7\times10^{-11}a^2\Omega_a$, with no pressure, $R_p = 0.5a$ and $J_2 = 0.01$. With $N_r = 100$ streamlines ($\mathcal{R} \simeq 0.09$ in the ringlet's middle, where the taper makes σ 1.25 times and Δ 0.8 times their ringlet-wide means) the ringlet stays smooth for 10^4 orbits while self-gravity, rather than J_2, controls its shape: within about 1000 orbits it straightens the initial twist, and the ringlet's body then stays nearly aligned, as two sub-ringlets that precess slowly relative to each other. With $N_r = 300$ ($\mathcal{R} \simeq 0.3$) and $N_r = 1000$ ($\mathcal{R} \simeq 0.9$) it becomes ragged and its streamlines cross after about 190 and 30 orbits. At $N_r = 1000$ the disturbance starts in the ringlet's interior, grows at the same rate with 40 or 241 particles per streamline, and does not grow when ring gravity is turned off, consistent with the self-gravity-assisted overstability for $\nu_b = \nu_s$ at $\mathcal{R} > 2/7$. Pressure with $Q \leq 2$ delays the crossing by less than two orbits, as expected from Figure 2b, where $\omega_p^2 \leq 0.1\kappa^2$.

The crossings near $\mathcal{R} \simeq 0.3$ are mostly a different, numerical, problem. The same ringlet without the initial twist, at $N_r = 333$ ($\mathcal{R} \simeq 0.31$), crosses after 145 orbits, and its code's own linearized axisymmetric modes, computed from the reference implementation's accelerations, grow by only 3.3×10^{-3} per orbit, far too slowly. The disturbance that crosses is instead a sawtooth along each streamline with azimuthal wavenumber $m \simeq N_t/2$, where N_t is the number of particles per streamline. It also grows in a circular ringlet, without ring gravity, and without viscosity, though more slowly (after about 220, 170 and 850 orbits), its onset varies non-monotonically with N_t (with $N_t = 30$ or 50 the ringlet survives 10^4 orbits, with 20, 40 or 320 it does not), and pressure with $Q \leq 10$ delays it by at most 20%. It is presumably seeded by the 3-point Lagrange interpolation in longitude that couples neighbouring streamlines' particles, and it remains an open problem.

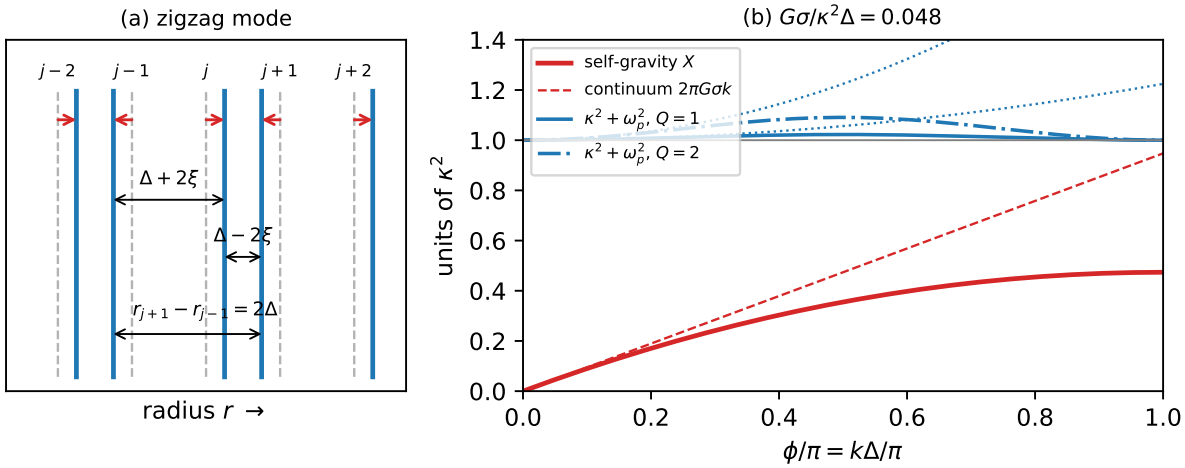


Figure 2. (a) The zigzag mode of a lattice of streamlines (blue) displaced by $\pm\xi$ from their unperturbed radii (dashed): the gaps alternate between $\Delta \pm 2\xi$, but $r_{j+1} - r_{j-1} = 2\Delta$ does not change, so neither does the surface density that sets streamline j's pressure. (b) The terms of Eqn. (B5) versus $\phi = k\Delta$, in units of κ^2, for $G\sigma/\kappa^2\Delta = 0.048$ (the ringlet of the text with $N_r = 1000$): the destabilizing self-gravity X (Eqn. B3, solid red) and its continuum value $2\pi G\sigma k$ (dashed red), and the restoring $\kappa^2 + \omega_p^2$ for $Q = 1$ and 2 (blue; Eqn. B4) and their continuum values $\kappa^2 + c^2k^2$ (dotted blue). A mode is unstable where X exceeds $\kappa^2 + \omega_p^2$.

So Eqn. (B6) is a necessary condition only. In practice a simulated ringlet whose self-gravity matters should be run with $\mathcal{R} \lesssim 0.1$, viscous or not, and a ringlet that needs more streamlines must be made narrower and much less massive.

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